TECHNICAL ARCHITECTURE BRIEF · FOR EXTERNAL REVIEW

Vision-led agentic survey automation on Android (TopSurveys / Qualtrics WebView)

Repository: github.com/suvmitra/lion-opal-rose-moss · Package: survey-agent · Engine version: **24.0.115**

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This document describes what the program is, how it is built, how a single agent step runs, how answers are chosen, how taps are grounded on a phone screenshot, where large language models are used (and where they are not), and the failure modes that still dominate live runs. It is written so an independent expert can recommend the next architectural refinement without having to reverse-engineer 15,700 lines of concatenated Python.

1. Executive summary

The agent fills paid mobile surveys inside the TopSurveys Android app. The survey UI is a Qualtrics (or similar) form rendered in a WebView, inside Aloha/Chrome-like chrome, on a physical Samsung-class device driven over ADB. There is no official survey API. Perception is a screenshot + Tesseract OCR + UIAutomator node dump. Action is `adb shell input tap/swipe/text` and a WebView click injection.

Control is a deterministic router of specialised screen handlers, not an unconstrained LLM loop. Gemini (and optionally Salad-hosted Qwen-VL) are backups when local routers cannot classify the screen. A knowledge base and persona file supply declared facts (age 44, Male, Dublin, child_age 8, …). Missing personal facts can be typed on the operator terminal and persisted.

- **Goal:** complete a survey as a consistent human persona, as fast as is reliable, without minimizing the app or tapping the Android nav bar.
- **Loop:** observe → classify phase → specialised router → inspect/critique → execute → verify postcondition → learn (RL / screen signature).
- **Hard constraint:** each engine part file ≤ 800 lines; loaded into one namespace by bootstrap.py.
- **Current version:** 24.0.115 (green Participate pill center, mindset critique, county Dublin-then-scroll).
- **Dominant live failure (as of 2026-09-21):** operator still launching v24.0.93 from `clinical_trials\survey_automation` instead of this repo; OCR baseline y=899 mapped to native (718,2776) and was refused as the system nav.

2. Problem the software is solving

Market-research surveys on this channel are multi-page Qualtrics instruments: radios, checkboxes, dropdowns, free text, ranking, Likert sliders, MaxDiff most/least columns, matrices, captchas, drag-and-drop, cookie overlays, and sticky Continue footers. The WebView does not expose a stable DOM to the automation host. UIAutomator XML is incomplete for Qualtrics widgets (editable=0 on many screens). Therefore the program treats the screenshot as the source of truth and accessibility nodes as a hint.

In-scope

- Pick a Featured Survey tile from the TopSurveys dashboard.
- Pass landing CTAs (Participate / Start!), cookie banners, I Agree.
- Answer questions using knowledge_base.json and persona_data.py.
- Ask the human on the terminal only when a personal fact is missing and inference is not allowed.
- Scroll until Continue/Next is visible; tap it once the answer is filled.
- Recover from validation banners, keyboards covering Continue, and unchanged screens.

Out of scope / policy limits

- No medical, legal, financial, credential, or security inference (restricted categories in the KB).
- Camera / microphone / location permissions default deny (PERMISSION_POLICY=safe_only).
- Document picker may only select AUTHORIZED_UPLOAD_FILE.
- Does not drive the survey vendor's website in a desktop browser; Android only.
- Does not claim 100% survey completion; MAX_AGENT_STEPS=200 and a stagnation halt exist.

3. Runtime environment

Layer	Choice	Notes
Host OS	Windows 10/11 (operator)	Developed and run under E:\AIML\dev\lion-opal-rose-moss\survey-agent
Python	3.12 venv	Dependencies: opencv-python, pytesseract, pillow, imagehash, requests, python-dotenv
Device	Physical Android phone via USB	Typical screenshot 1440×3088; ADB default C:\adb\adb.exe
App under test	topsurveys.paid.survey.money.cash	Surveys render in a WebView (Qualtrics-like)
OCR	Tesseract	Full-page + crop PSM 7; weak on white-on-orange and status-bar noise
Local VLM	Salad Cloud Qwen2.5-VL-7B (optional)	Often 404/503; marked offline for the rest of the run
Cloud VLM	Gemini 3.5 Flash-Lite then 3.6 Flash	OpenAI-compatible endpoint; hourly cap default $0.30
Source of truth repo	github.com/suvmitra/lion-opal-rose-moss	Branch main; do not run the old clinical_trials copy

Table 1. Deployment stack.

4. Repository and load model

survey_with_salad.py is a 22-line entry point. It calls survey_engine.bootstrap.load_engine(globals()), which exec's every file in survey_engine/parts/ in numeric order into one shared namespace. There are no circular imports because there are no intra-package imports: later files simply call functions defined earlier. bootstrap refuses to load a part that exceeds 800 lines.

```
survey-agent/
  survey_with_salad.py          # entry: load_engine + main()
  human_answer_matching.py      # terminal option matching
  knowledge_base.json           # declared facts + RL policy blob
  persona_data.py               # richer persona narrative
  survey_engine/
    bootstrap.py                # concat parts into one namespace
    parts/00_config.py … 26_reason.py
  archive/                      # monolith snapshots v24.0.16–0.87
  fixtures/                     # regression screenshots
```

5. Module map (parts)

Part	Lines	Responsibility
00_config	129	VERSION, env, paths, logger, ADB/Tesseract/Gemini/Salad knobs
01_changelog	624	Per-version change bullets used at startup
02_runtime	505	Observation/ActionResult, fresh-run cleanup, ADB, screenshot, OCR, UI dump
03_labels	483	CTA lexicons, landing/consent/dashboard detectors
04_rank_likert	530	Ranking, Likert, classify_screen_phase

Part	Lines	Responsibility
05_kb_match	651	Tiles, option matching, KB load/write
06_learning	765	Lessons, learned-screen signatures, tabular RL
07_vlm	793	Salad + Gemini vision, stagnation VLM, failed-tap memory
08_landing	792	milli→native, Participate/Start, I Agree, orange/green CTA, terminal close
09_county	720	Irish county list: Dublin from KB, orange row, footer Continue
10_children	437	Children count radios + child_age text
11_carousel	480	Horizontal carousel rows
12_maxdiff_selectall	635	MaxDiff most/least + select-all checklists
13_select_one	269	Single-choice radios + select_then_continue
14_matrix	725	Matrix grids, visual_progression_coords
15_progression	550	Continue/Next, prefer_progression_over_repeat
16_inspect	800	Last-chance inspector (gender, county, banners) + critique hook
17_human	562	Terminal key-in; writes KB
18–23 checks	~2.4k	Regression self-checks (dashboard, county, age, RL, MaxDiff, …)
20_text_fields / 21_open_text	1.0k	Age/postcode geometry, captcha, drag-drop, comments
22_execute	777	Tap/type/scroll, SAFE TAP, postcondition verify
24_preflight	366	Runs self-checks then Salad/Gemini/device probes
25_runner	800	observe() + main() agent loop
26_reason	258	Mindset prompt, green_cta_milli, critique_decision, THINK/VLM

Table 2. Engine parts. Total ~15,768 lines including the entry file.

6. The agent loop

main() in 25_runner.py is a bounded for-loop (default 200 steps). Each step is one Observe–Decide–Inspect–Act–Verify cycle. initialize_fresh_run() deletes screenshots, truncates debug_runner.log and automation_log.csv, and resets agent_state.json. Persistent knowledge_base.json is retained.

6.1 Observe

- ensure_survey_foreground() — relaunch TopSurveys if the app was sent home.
- capture_screenshot() — PNG + sha256 + perceptual hash; identical phash reuses the previous file.
- ocr() — Tesseract on the full screenshot.
- dump_ui_nodes() — UIAutomator dump; often 0 editable nodes on Qualtrics pages.
- classify_screen_phase() — dashboard | consent_overlay | survey_landing | survey_question | permission_dialog | document_picker | loading | completion | terminal_modal | navigation.
- Specialist detectors can force phase=survey_question (MaxDiff, matrix, children, county, …).
- load_kb() always; load_persona() on survey_question.

6.2 Decide (router priority)

Hard routes run first so a specialised screen cannot be stolen by RL or Gemini:

```
1. I Agree / Decline consent
2. Gender cards (Are you? Male/Female)
3. Adult age text box (KB age=44)
4. If screen unchanged N times → stagnation_gate → Gemini snapshot JSON
5. Widget specialists (order matters):
     close-confirm, Likert, ranking, drag-drop, captcha, select-all,
```

```
     I Agree, select-one, open text, county list, children, child ages,
     carousel, MaxDiff, occupation group, matrix
6. Phase specialists: dashboard tile, permission, picker, cookie overlay,
     landing Participate/Start
7. On survey_question leftover:
     lessons → RL replay → generalized_unseen_router → exact KB option →
     declared preference → inferred preference → text KB → human prompt →
     learned-screen action → Gemini observation prompt
8. Post-decision repairs: text payload copy, human if empty personal,
     authorized fallback, prefer_progression_over_repeat, inspect_decision
     (which calls critique_decision), payload-None guard on select-one
```

6.3 Act and verify

execute_action() maps milli coordinates [0–1000] to device pixels, refuses or clamps taps that would hit the Android navigation bar, types via keyevents or ADB text, scrolls with a swipe (list_scroll uses a shorter inner-list swipe). After the action, observe(light) and verify_postcondition() check hashes, OCR, orange/green bars, and widget-specific predicates (e.g. county list still open after tapping Dublin is success, not failure).

Reward is written into a small tabular RL store in knowledge_base.json and a per-signature learned_screen_actions table. Failures go to _FAILED_TAPS so the next loop must not repeat the same (x,y).

7. Coordinate system (critical)

All decisions speak milli-coordinates: integers 0–1000 on each axis of the screenshot. Execution converts with normalized_to_native():

```
sx = image_width  * x / 1000
sy = image_height * y / 1000
if device_size != image_size:
    sx *= device_width  / image_width
    sy *= device_height / image_height
```

On the operator phone a screenshot is typically 1440×3088 and includes the status bar and the Android 3-button nav. That means:

Milli y	Approx. native y (3088)	What lives there
0–80	0–247	Status bar / Aloha X and title
280–520	864–1605	Question body, gender cards, age field, county rows
770–875	2378–2702	In-content Continue / selected list row (Wexford, Kerry)
850–875	2625–2702	Intended Participate / footer Continue CENTER
890–920	2748–2841	OCR text baseline of a bottom pill; often the nav
>940	>2902	Android system nav — SAFE TAP must never land here

Table 3. Empirical y-bands on a 1440×3088 screenshot.

Grounding methods, in intended order of trust: (1) HSV blob of the actual coloured control — orange_cta_milli for Qualtrics Continue, green_cta_milli for Participate (v24.0.115); (2) crop OCR of a known label; (3) accessibility node bounds, after discarding nav-bar and chrome nodes; (4) geometric fallbacks such as [500,860]. Tesseract's word box is the baseline of the glyphs, which sits below the visual centre of a pill. That single fact caused the 20+ iteration Participate stall: OCR y=899 → native 2776 → SAFE TAP refuse → Gemini repeats 900 → local router overwrites back to 899.

8. Perception stack

Signal	Producer	Used for	Failure mode
PNG screenshot	ADB screencap	VLM, HSV, OCR, phash	Dedup can hide a small orange change
phash	imagehash	stagnation, screenshot reuse	Wexford gray↔orange can still 'change'
OCR text	Tesseract	phase, question, options	Misses white-on-orange 'Dublin'; reads status bar

Signal	Producer	Used for	Failure mode
UIAutomator XML	uiautomator dump	clickable nodes, bounds	Qualtrics fields often absent / y=92% chrome
HSV orange	OpenCV inRange H 5–25	Continue footer vs selected row	Selected county row is also orange
HSV green	OpenCV H 35–95 (v115)	Participate / Start pill centre	Needs live screenshot; tests have none
Crop OCR PSM 7	Tesseract on ROI	age digits in the box	Red 'no more than 99' banner leaked as 99

Table 4. Sensors. Screenshot is authoritative; XML is a hint.

9. Language models — where they sit

The design intent is: local geometry and the knowledge base answer known screens; the VLM is asked only when the router has no specialist or the screen is stagnating. In practice Gemini is still invoked too often because (a) Salad is frequently offline, (b) stagnation fires after 1–4 unchanged cycles, (c) inspect/landing routers overwrite the VLM JSON with the same OCR coordinate that already failed.

Role	Model / stack	When used
Primary VLM (config)	Salad Qwen2.5-VL-7B via SALAD_URL	If preflight 200; else marked offline for the run
Fallback VLM	gemini-3.5-flash-lite then gemini-3.6-flash	Observation prompt + stagnation JSON
Prompt wrapper	build_observation_prompt + mindset_prompt + ASK=	Phase task, OCR, compact UI, KB/persona
Stagnation JSON	stagnation_vlm_decision	Unchanged screen; must not reuse failed coords
Age THINK	think_from_screenshot	Read digits in the age box when Tesseract is wrong
Not used as planner	—	No multi-step tool-calling agent; one JSON action per call

Table 5. Model usage. Hourly spend cap VLM_HOURLY_BUDGET_USD (default 0.30).

VLM_PRIMARY defaults to salad. If Salad returns 404/503, mark_salad_offline() stops retrying Salad for the rest of the process. Gemini stays in use. The operator is not currently blocked when Salad is down, by design.

10. Knowledge, persona, and human input

knowledge_base.json holds declared facts the agent is allowed to type or tap without asking. persona_data.py is a richer narrative used as secondary evidence. Writes go back to the JSON with provenance (declared | inferred | human) and a confidence.

kb_key	Value	Typical widget
age	44	Text box + Continue
gender	Male	Cards or radios, then Continue
city / county	Dublin	Open list, tap row, scroll to footer Continue
area_code / postal_code	D22 / D22V048	Text
child_age / number_of_children	8 / 1	Text / radio
employment_status	Employed Full-Time	Select-one
smoking_status	No, I don't smoke	Select-one
household_pets	Cat(s), Dog(s), Fish	Multi-select
seniority / occupation	Senior Lead / middle manager	Occupational group list

Table 6. Core declared facts (see knowledge_base.json).

If a question is classified personal, the KB/persona miss, and inference is forbidden or low-confidence, capture_human_answer_decision() prints the question and visible options on the terminal. The operator may type labels ('Dublin'), indices ('3'), or 'skip'. Matching lives in human_answer_matching.py. A previous bug treated instructional paragraphs as options; matching now prefers numbered real choices and can accept 'option 3' / 'middle manager'.

11. Widget specialists

Widget	Strategy	Continue rule
Select-one	KB/persona/inference → tap radio (payload required)	select_then_continue or later survey_next
Multi-select	Tap remaining KB items; do not toggle off	When remaining=[]
Gender cards	Male at ~[280,280]; Continue ~[500,455]	Never re-tap Male (deselects)
Age text	Field ~55 milli above orange Continue; type 44 once	Dismiss keyboard, Continue; never 4444
County list	Open if closed; tap Dublin; orange between Donegal & Galway	Only footer y≥860; else scroll_down only
Children / ages	KB number_of_children / child_age=8	Type then Continue; do not prompt if KB has 8
Ranking	Tap remaining gray bars to assign 1..n	All items teal before Continue
Likert	Named scale → 3/4/5; teal handle = filled	No leftover up-arrow
MaxDiff	Most Important cell then Least Important cell	Then scroll/Continue
Select-all / claims	Check KB-relevant boxes, skip Other	Footer Continue after min checks
Matrix	Row×column from persona; one cell per row	Scroll until Continue
Open text	≥5-word human-like sentence, not 'AI generated'	Hide keyboard, Continue once
Captcha	Type on-screen distorted code	Next, not Continue
Drag-drop	Long-press tile, drop on dashed target	Then Next
Landing	green_cta_milli centre, WebView click	Must not hit nav y>0.92

Table 7. Specialist policies. Each has a session flag to prevent toggle/duplicate type.

12. Learning mechanisms

Three overlapping memories exist. They have helped on repeated screens and also caused regressions when a coordinate was rewarded for the wrong reason (e.g. tapping Wexford y=780 'changed' the screenshot because the row toggled orange, RL q rose, later replay hit Wexford instead of Continue).

- **learned_screen_actions** — signature from OCR keywords; reuse action if confidence ≥ LEARNED_ACTION_MIN_CONFIDENCE (0.80).
- **rl_policy** in knowledge_base.json — state = widget|question-stem, action_id = action|payload, tabular Q with n.
- **lessons / improve_survey_intelligence** — success/failure counters per semantic key.
- **generalized_unseen_router** — assume select-one / text / continue from visible controls when no specialist fires.
- Session flags (_COUNTY_SESSION, _AGE_SESSION, _I_AGREE_SESSION, …) are process-local and reset on fresh run; they must not leak across self-checks.

13. Mindset critique (v24.0.114+)

The operator asked that curiosity, rationality, open-mindedness, objectivity, empiricism, critical thinking, skepticism and intellectual honesty be implemented as behaviour, not slogans. 26_reason.py does this:

Trait	Code behaviour
Curiosity	Log SEE= first 120 OCR chars before every inspect

Trait	Code behaviour
Rationality	If landing shows Participate, force that CTA
Open-mind	If (x,y) is in failed_coords_set, retry y-40
Objectivity	Prefer green HSV blob over remembered/OCR y
Empiricism	Unchanged phash ⇒ last tap missed; do not repeat it
Critical thinking	Is Continue/Participate actually on this screenshot?
Skepticism	y≥880 Participate → clamp 860; y≥940 is Android nav
Honesty	Continue with payload=None on a radio page is rejected; fill first

Table 8. Mindset is executed in critique_decision(), called from inspect_decision().

mindset_prompt() is also prepended to Gemini observation and stagnation prompts so the image model is asked to apply the same rules.

14. Execution safety

- SAFE TAP: native y ≥ nav_ratio×height is refused, except Participate which is clamped to 0.860×height and WebView-clicked.
- Forbidden targets: privacy-policy footer, Android recents/home/back, cookie 'X' when a real CTA exists, instruction paragraphs.
- prefer_progression_over_repeat: if Male (or any option) was already issued, do not tap it again (Qualtrics toggles cards off).
- Foreground watchdog: if TopSurveys is not resumed, relaunch the activity (a nav-bar miss sends the app home).
- Postcondition examples: 'screen changed', 'selected option is Dublin', 'field contains 44', 'maxdiff most selected'.

15. Quality gates

preflight() runs a battery of Observation-built self-checks before ADB or Gemini are used in anger. A failed check aborts the process with a diff of expected vs actual decision dicts. That is why many operator logs start with a wall of [SELF-CHECK] lines — those are not live taps.

- Dashboard tile selection, cookie overlay, Participate landing, Start! vs Continue.
- I Agree then orange Continue (not nav-bar).
- Postcode D22, smoking, pets multi-select, gender Continue, age 44 payload repair.
- County Dublin-not-Donegal, open-list tap, past-list scroll_up only before Dublin is chosen.
- RL select_one replay, MaxDiff most then least then Continue, ranking, Likert.
- Mindset: Participate y=899 clamped into 850–875.
- Then: Salad probe, Gemini probe, adb devices.

16. Known failure modes (honest)

The system is a growing collection of geometric specialists around a screenshot. It is not yet a general GUI agent. The following are the failures an external reviewer should treat as first-class, not as one-off bugs.

ID	Symptom	Structural cause	Current mitigation
F1	20+ loops on Participate	OCR baseline ≠ pill centre; SAFE TAP refuse; VLM overwritten	green_cta_milli + clamp (115). Operator must run this repo.
F2	Male selected then unselected	Second tap toggles Qualtrics card	prefer_progression_over_repeat; gender session
F3	Age 4444 / banner 99	Type twice; crop OCR reads error	typed flag, crop ignore banner, THINK
F4	County Kerry/Wexford as Continue	Any orange row treated as CTA; y=780	Lowest orange y≥860 only; Dublin alignment

ID	Symptom	Structural cause	Current mitigation
F5	County up/down oscillation	OCR misses white-on-orange Dublin	tapped/picked ⇒ scroll_down only
F6	Continue while unanswered	Progression repair too eager	required_answer_missing veto
F7	Human prompt storm	Inspector reprints missing_fact_key	Once per fact key; reload KB from disk
F8	Wrong folder / old version	Two checkouts; screenshots path in clinical_trials	Log CODE DIR; this PDF
F9	RL false-positive reward	Toggle changes phash, q↑	Do not replay y<860 on county lists
F10	Gemini cost / latency	Stagnation VLM every unchanged cycle	Need: stop calling VLM with the same failed coord
F11	Salad 404/503	Container not up	Offline for run; Gemini continues
F12	Module split at 800 lines	inspect/runner/vlm at the cap	Hard to add logic without deleting lines

Table 9. Reviewer punch-list. F1 and F8 are operational; F10 is architectural.

16b. Core types

Every step is an Observation in, a decision dict out, an ActionResult after execute.

```
@dataclass
class Observation:
    step: int
    screenshot: Optional[str]      # PNG path on the host
    image_width, image_height: int # screenshot pixels (often 1440x3088)
    device_width, device_height: int
    screen_hash: Optional[str]     # sha256; phash stored alongside
    ocr_text: str
    ui_nodes: list[dict]           # UIAutomator {id,text,bounds,clickable,...}
    kb: dict                       # knowledge_base.json
    persona: dict                  # persona_data.MASTER_USER_PERSONA
    screen_phase: str
    survey_tiles: list[dict]       # dashboard cards only

@dataclass
class ActionResult:
    success: bool
    changed: bool                  # hash or OCR differed
    message: str
    after_hash: Optional[str]
```

17. Decision object (contract)

```
{
  "screen_state": "survey_question|survey_landing|dashboard|...",
  "question_text": str,
  "action": "click|select_option|select_then_continue|type_text|\n           scroll_down|scroll_up|wait|finish|request_hu
  "target_element_id": int|None,     # UIAutomator node id
  "target_text": str|None,           # label to tap or type
  "target_coords": [x_milli, y_milli],
  "coordinate_authoritative": bool,  # screenshot beat the node
  "input_payload": str|None,         # must be set for radios/text
  "kb_key": str|None, "kb_write": bool,
  "confidence_score": 0..1,
  "postcondition": "screen changed|selected option is Dublin|...",
  "answer_reason": str, "visual_reasoning": str,
  "answer_provenance": "verified|declared|inferred|human"
}
```

18. How to run (operator)

```
cd E:\AIML\dev\lion-opal-rose-moss
git pull origin main
```

```
cd survey-agent
# .venv already created; .env holds GEMINI_API_KEY, optional SALAD_*, ADB_PATH
python survey_with_salad.py

First log lines MUST contain:
  FRESH SURVEY AUTOMATION SESSION - v24.0.115
  CODE DIR=...\lion-opal-rose-moss\survey-agent
If you see v24.0.93 or clinical_trials\survey_automation you are on the old copy.
```

Artifacts per run: screenshots/step_NNNN.png, _verify_*.png, debug_runner.log, automation_log.csv, agent_state.json. knowledge_base.json is the only file intended to accumulate across runs.

19. Questions for the consultant

The operator will use your recommendations for the next refinement. Please be opinionated on the following.

Architecture

- Should the 27-file exec-into-one-namespace model be replaced by a real package with typed interfaces, even if files stay small?
- Is a general GUI agent (screenshot → VLM → grounded SAM/UI-TARS/OmniParser boxes) a better backbone than per-widget routers, given the $0.30/hour cap?
- How should perception be unified: AccessibilityNode + screenshot crop embeddings + a single 'control detector' instead of orange HSV, green HSV, OCR, and XML fighting?
- What is the right coordinate frame — screenshot milli, dp, or a nav-bar-subtracted content box?

Control and learning

- The inspect + prefer_progression + RL replay stack often undoes a correct specialist. Propose a single authority (policy head) with explicit vetoes.
- Tabular RL rewarded a toggle as success. What learning signal is safe on a GUI where many taps are reversible?
- Stagnation currently re-asks Gemini every cycle. Propose a budgeted planner: one VLM sketch per unique phash, then only local retries.

Reliability

- How would you test this without a phone — screenshot replay, recorded ADB, or a Qualtrics fixture app?
- What SLOs are reasonable (median seconds per question, Continue success rate, Gemini $ / survey)?
- Is human-in-the-loop on the terminal acceptable in production, or should missing facts be collected in a pre-survey interview?

20. Version lineage (selected)

The product evolved by live-screenshot patches, v24.0.16 through v24.0.115. Monolith snapshots live in archive/ up to 0.87; after that the engine is split. Highlights a reviewer may care about:

Ver	Theme
0.93	I Agree OCR; TypeError int(list) on Continue coords — this is the copy still running in clinical_trials
0.98–0.99	Gender cards: tap Male once, then mid-screen Continue
0.100–0.106	Age 44 once, not 4444; field above Continue; keyboard dismiss
0.107–0.112	County: Dublin orange, never Kerry/Wexford as Continue; scroll down only after pick
0.113–0.115	Participate pill centre, SAFE TAP clamp, mindset critique, green HSV blob, CODE DIR log

Table 10. Recent versions. Always confirm the live log header, not git describe on another folder.

21. Appendix — important environment variables

Variable	Default	Meaning
ADB_PATH / DEVICE_ID	C:\adb\adb.exe / first device	Phone bridge
TESSERACT_CMD	C:\Program Files\Tesseract-OCR\tesseract.exe	OCR binary
GEMINI_API_KEY	(required if Salad down)	Google AI Studio key
SALAD_URL / SALAD_API_KEY	empty	Optional local VL
VLM_PRIMARY	salad	salad | gemini
VLM_HOURLY_BUDGET_USD	0.30	Hard-ish spend cap
MAX_AGENT_STEPS	200	Loop ceiling
MAX_STAGNATION_CYCLES	4	Unchanged-screen halt
ALLOW_INTERACTIVE_HUMAN_INPUT	true	Terminal prompts
ALLOW_PERSONAL_INFERENCE	true	Non-restricted guesses
SURVEY_SELECTION_POLICY	highest_value	Dashboard tile pick
COOKIE_CONSENT_ACTION	accept	accept | reject
PERMISSION_POLICY	safe_only	Android permission dialogs
KB_FILE / PERSONA_FILE	next to the entry script	Facts

Table 11. Configuration. Loaded from survey-agent/.env.

22. Appendix — one-step sequence diagram (textual)

```
Phone --screencap/ocr/dump--> observe()
observe() --> classify_screen_phase + load_kb
          --> specialist routers (priority list §6.2)
          --> optional Gemini JSON if none / stagnating
          --> repairs + inspect_decision
                  \--> critique_decision (mindset, green blob, failed coords)
          --> execute_action (milli→native, SAFE TAP, webview or adb)
Phone <-- tap/swipe/type --+
          --> verify_postcondition (hash, OCR, orange/green)
          --> RL / learned-screen / lessons update
          --> next step or stagnation halt
```

23. Latency, cost, and ethics (for the reviewer)

A healthy step (no VLM) is typically 3–8 seconds: screencap ~1s, Tesseract ~2–4s, UI dump ~1–3s, tap ~0.3s, verify screencap. A stagnation cycle that calls Gemini 3.6 Flash adds ~5s and about $0.0018. Twenty refused Participate taps therefore burned ~$0.04 and ~two minutes on a button that a human would tap once. The $0.30/hour cap is an operator constraint; architecture that re-asks the VLM on the same phash violates it in spirit even when the dollar cap is not hit.

This software automates a paid consumer survey app using a declared persona. An external reviewer should flag terms-of-service, survey integrity, and personal-data handling (knowledge_base.json contains a real name, employer, child age). Those issues are product/legal, not solved in code. The technical brief assumes the operator is accountable for use.

Suggested reading order for the consultant: this PDF; survey_engine/bootstrap.py; parts/25_runner.py main(); parts/22_execute.py SAFE TAP; parts/26_reason.py; parts/09_county.py; one live debug_runner.log + matching step_*.png.

24. Document control

This brief describes engine **v24.0.115** as pushed to github.com/suvmitra/lion-opal-rose-moss (commit on branch main, generated 2026-09-21 09:47 UTC). It is not a specification of a future system. Where the code and this text disagree, the code in survey_engine/parts/ wins. Personal facts in knowledge_base.json (name, employer, child) are the operator's

declared survey persona and should be treated as confidential.

Prepared for independent architectural review. Next refinement should be driven by the consultant's answers in §19, not by another one-screen coordinate patch unless that patch is the recommended path.